UNIVERSITY OF INFORMATION TECHNOLOGY AND SCIENCES

Lab Test 2

Course Title:

Microprocessors, Microcontrollers and Assembly Language Lab

Course ID: CSE0611324

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Code:

```
.MODEL SMALL
.STACK 100H

.DATA
    MSG_B   DB 'Enter B: $'
    MSG_A   DB 0DH, 0AH, 'Enter A: $'
    MSG_MIN DB 0DH, 0AH, 'Smallest between C and D: $'

    A       DB ?
    B       DB ?
    C       DB ?    ; Quotient
    D       DB ?    ; Remainder
    MIN     DB ?    ; Minimum value

.CODE
MAIN PROC

    MOV AX, @DATA
    MOV DS, AX



    ;;

    LEA DX, MSG_B
    MOV AH, 09H
    INT 21H

    MOV AH, 01H     ; Read digit for B
    INT 21H
    SUB AL, '0'
    MOV B, AL
```

```
     LEA DX, MSG_A
    MOV AH, 09H
    INT 21H

    MOV AH, 01H      ; Read digit for A
    INT 21H
    SUB AL, '0'
    MOV A, AL



    ; New line after inputs

    MOV AH, 02H
    MOV DL, 0DH
    INT 21H
    MOV DL, 0AH
    INT 21H


    MOV AL, A
    XOR AH, AH  ; here divided program written
    DIV B

    MOV C, AL
    MOV D, AH




OUTER_LOOP:

    XOR CX, CX
    MOV CL, A
    MOV BX, 1        ; Initial count for '%' in first row

ROW_LOOP:
    PUSH CX          ; save the Outer loop counter in Stack
    MOV CX, BX
```

```
PERCENT_LOOP:
    MOV AH, 02H
    MOV DL, '%'
    INT 21H
    LOOP PERCENT_LOOP

    ; New line print
    MOV AH, 02H
    MOV DL, 0DH
    INT 21H
    MOV DL, 0AH
    INT 21H

    INC BX                          ; 1 % will be increment
    POP CX
    LOOP ROW_LOOP


    MOV AL, C
    CMP AL, D
    JBE C_IS_SMALL
    MOV AL, D       ;here, if D is small then AL will be = D



C_IS_SMALL:
    MOV MIN, AL


  ; Print smallest number



    LEA DX, MSG_MIN
    MOV AH, 09H
    INT 21H
```

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```
    MOV DL, MIN
    ADD DL, '0'      ; Convert number back to ASCII code
    MOV AH, 02H
    INT 21H

    ; Exit program

    MOV AH, 4CH
    INT 21H          ; this code for exit from program and run

MAIN ENDP
END MAIN
```

Output Screenshot:

